LAKE SUPERIOR

Ice-out on the Big Lake

Lake Superior's winter ice is arriving later and leaving sooner. Ports, anglers and shoreline towns are learning to plan around it.

By **Owen Brandt**
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DULUTH, Minn. - The ice left the harbor this spring the way it has in most recent years: early, quietly and without much of a fight.

There was no last stand at the ship canal, no Coast Guard cutter grinding a path for a line of waiting freighters, no crowd on the pier at Canal Park to cheer the first boat through. The gray sheet that had covered the inner harbor since late January went soft in the second week of March. A cutter made one last icebreaking pass along the main channel on March 17. Two mild nights and a southwest wind later, the water under the Aerial Lift Bridge was open and black, and the gulls had the breakwater to themselves.

"Not that long ago, we'd still be fighting ice in here at the end of April," said a harbor pilot who has worked the port for three decades. "This year the only thing I was waiting on was the calendar."

For the people who work the big lake, ice is a calendar. It decides when the shipping season can open, when a charter captain can put a boat in, and how much of the shoreline the winter storms can reach. A shorter ice season moves all of those dates at once.

This winter's ice season was short even by recent standards, and it ended in a hurry.

A quick exit

The most detailed account of Superior's ice comes from the daily ice charts kept by NOAA's Great Lakes Environmental Research Laboratory in Ann Arbor, Michigan. The charts are built from satellite images and ice analysts' observations, and the lab's record goes back to 1973.

In an average winter in that record, ice covers about 60% of Superior at its peak, usually in late February or early March.

This winter never came close. A cold stretch in late January and early February pushed the ice to its high for the season, 38% of the lake, on Feb. 10. Then the cold broke. By March 1, ice covered 24% of Superior. By March 25, the day the Soo Locks opened for the season, it covered 6%. On April 1, the lab's chart showed about 2%, most of it tucked into a few sheltered bays on the Canadian shore.

“It’s not that the ice never formed,” said an ice analyst at the Ann Arbor lab. “It formed late, it never got very thick, and when March turned warm there wasn’t much holding it together.”

Superior’s short ice season

Peak ice cover this winter	**38% of the lake, Feb. 10**
Average peak ice cover, 1973 to 2025	**About 60%**
Ice cover on March 25, when the Soo Locks opened	**6%**
Ice cover on April 1	**About 2%**
Last icebreaking run, Duluth-Superior harbor	**March 17**
First oceangoing ship of the season in Duluth	**April 3**

Sources: NOAA Great Lakes Environmental Research Laboratory daily ice charts; U.S. Coast Guard; Duluth Seaway Port Authority.

The shipping calendar

The shipping season runs on a schedule the ice doesn’t set. The Soo Locks at Sault Ste. Marie, Michigan, the only way for big freighters to travel between Superior and the lower lakes, close on Jan. 15 and reopen on March 25 every year so the U.S. Army Corps of Engineers can work on them. The St. Lawrence Seaway, the route in from the Atlantic for oceangoing ships, opens in late March.

What the ice decides is how hard and how costly the first weeks are. In a hard winter, freighters leave their winter berths into harbor ice two or three feet thick. Coast Guard icebreakers work for weeks to keep the channels open, and a loaded ship can spend a day covering water it would cross in an hour in July. Ice scrapes paint from hulls and dents steel. Schedules slip, and so do the cargoes that ride on them, from iron ore pellets bound for steel mills to grain bound for export.

This spring, none of that happened. The freighters that spent the winter tied up in the Twin Ports left through open water. The first oceangoing ship of the season, a “saltie” in waterfront talk, came under the Aerial Lift Bridge on April 3 without meeting ice anywhere between the Soo and the docks.

For a pilot, an open harbor in March is easy work. It is also a change you can feel from the wheelhouse. The pilot put it simply: the lake used to take the winter off, and now it mostly doesn’t.

A century of notebooks

The satellite charts tell the story of one winter in fine detail. For the longer story, researchers turn to older and humbler records.

In a few harbors and bays around Superior, people have been writing down the date the ice went out for more than a century: in lighthouse logs, in harbor masters’ ledgers and in the newspaper notices that announced the first boat of spring.

Some of the most useful records are kept by volunteers. Up and down the shore, people with a view of a bay write down, every spring, the day the ice goes out. A few have written that date in the same notebook for 40 years. They don't measure anything. They look out the window and make a call. But the same person making the same call from the same window, year after year, produces a record that researchers prize. It reaches back before the satellites, and it shows what one bay did while the charts show the whole lake.

Read side by side, the old records and the new ones point the same way. In most of those harbors and bays, the ice now goes out earlier than it did a century ago, and winters when the bay barely froze, once unusual, have become ordinary.

Researchers who track ice cover caution that the swings from one winter to the next are large, and that one warm year proves little. The longer record is clearer. Ice is forming later and breaking up earlier, and open water in January no longer surprises anyone on the waterfront.

Fishing on the lake's schedule

Out of Knife River, about 20 miles up the shore from Duluth, the early open water put boats on the lake weeks ahead of a normal spring.

The charter captains went first. The weeks just after the ice leaves are some of the best lake trout fishing of the year, when the fish feed near the surface in water that is still close to freezing. This year those weeks began in March.

The ice anglers had the opposite spring. On the North Shore, where safe ice on the big lake is rare in any winter, the season depends on a few sheltered harbors and bays. This year most of them were unsafe by the first week of March, and some never froze well enough to walk on.

A commercial fisherman who has set nets out of Knife River for more than 30 years was on the water in the first week of April, two or three weeks earlier than a normal spring allows. The catch is herring, the small, silver fish that once filled the North Shore's fish houses and still supports a small number of licensed commercial operations on the Minnesota side of the lake.

"An early start sounds like a gift, and some years it is," the fisherman said. "But the fish don't read the calendar. The herring were there. They were just spread out, and you burn a lot of fuel finding them."

The bigger question for the fishery is what happens under the ice, or without it. Herring spawn in late fall, and their eggs lie on the lake bottom through the winter. Fisheries researchers have reported that Superior's strongest crops of young herring tend to follow long, icy winters, possibly because ice shelters the eggs from storms that churn open water. The link is still debated, and plenty else matters, from the number of spawning fish to how much food the young find in spring. But a string of short ice seasons takes away one of the things that has tended to help.

A shore without its armor

Ice protects the shore, too. In a cold winter, frozen spray and broken floes build into a ridge along the beaches and rocky points, and the waves break against that instead of the land. On a mostly open lake, winter storms reach the shore directly.

From Duluth to Grand Marais, the people who look after harbors, parks and shoreline roads now talk about winter storms the way they once talked only about the gales of November. A storm that would have spent itself on shore ice can take out a stretch of trail, strip a beach to cobble or chew at the foot of a bluff, and the repairs come out of budgets built for calmer winters.

What one spring can and can't tell us

One early spring is weather. The harder question is what a run of them adds up to, and that is where the data start to run thin.

The satellite record is a little more than 50 years long, which is not much for a lake this size. Superior is so deep that it carries summer's heat well into winter, so a warm summer and fall can leave it slow to freeze no matter what January brings. Its ice also swings with patterns such as El Niño, which tends to bring milder winters to the upper Midwest.

The ice analyst said the long-term decline in Superior's ice holds up after those swings are taken into account. But no one can say from this winter's charts alone how much of this spring was a warming climate and how much was a warm March.

The older harbor records have limits of their own. A notebook tells you when the ice left one bay, not the whole lake, and a lighthouse keeper's "ice out" in 1910 may not mean exactly what a volunteer's does today. Researchers who use those records say so.

Nor can the records answer the questions people on the water ask most: whether herring will keep producing strong year classes with less ice, how many winters of open-water storms the shore can take, and whether a winter like 2014's, when ice covered nearly all of Superior, is now a rarity or simply overdue.

On that last question, the ice analyst was careful. "Cold winters haven't gone away," the analyst said. "They're farther apart, and the lake needs a longer, harder cold to freeze than it used to."

In the notebooks up and down the shore, this spring's date is already written in, below all the others and earlier than most. Next spring's line is still blank. Nobody on the waterfront would bet much on what goes there.

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